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Your Roll No.....

Sr. No. of Question Paper : 1567

F-8

Unique Paper Code : 2351401

Name of the Paper : ANALYSIS III

(Riemann Integration and Series of Functions)

Name of the Course : B.Sc. (H.) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt any two parts from each question.
3. All questions are compulsory.

1. (a) Let f be a bounded function on $[a, b]$. Show that $L(f) \leq U(f)$, where $L(f)$ is the lower Darboux integral and $U(f)$ is the upper Darboux integral on $[a, b]$. (6)

(b) Let f be a continuous function on $\mathbb{R}$ and define

$F(x) = \int_{x-1}^{x+1} f(t) dt$ for $x \in \mathbb{R}$. Show that F is differentiable function on $\mathbb{R}$. Compute F' . (6)

P.T.O.

$$(i) \int_1^{\infty} \frac{1}{x\sqrt{x+1}} dx$$

$$(ii) \int_1^{\infty} e^{t^2} dt$$

(3+3)

4. (a) Let $\langle f_n \rangle$ be a sequence of continuous functions on $A \subseteq \mathbb{R}$ and suppose that $\langle f_n \rangle$ converges uniformly on A to $f: A \rightarrow \mathbb{R}$. Then show that f is continuous on A . (6½)

- (b) Let $f_n(x) = \frac{nx}{1+n^2x^2}$ for $x \geq 0$. Show that the sequence $\langle f_n \rangle$ converges only pointwise on $[0, \infty[$ and converges uniformly on $[a, \infty[$, $a > 0$. (6½)

- (c) Let $f_n(x) = \frac{x}{1+nx^2}$ for $x \in \mathbb{R}$, $n \in \mathbb{N}$. Show that the sequence $\langle f_n \rangle$ converges uniformly on $\mathbb{R}$. (6½)

5. (a) State and prove Weierstrass-M test for the uniform convergence of a series of functions. (2,5)

- (b) Show that the series of function $\sum_{n=1}^{\infty} \sin\left(\frac{x}{n^2}\right)$ converges uniformly for $|x| \leq a$, $a > 0$ and is not uniformly convergent on $\mathbb{R}$. (7)

P.T.O.

- (c) Suppose f and g are continuous functions that $g(x) \geq 0$ for all $x \in [a, b]$. Prove there is c in $[a, b]$ such that:

$$\int_a^b f(t)g(t)dt = f(c) \int_a^b g(t)dt.$$

2. (a) Prove that every monotonic function is integrable.

- (b) Let f be defined on $[0, b]$ as $f(x) = \begin{cases} x, \\ 0, \end{cases}$

Calculate upper and lower Darboux integrals on $[0, b]$. Is f integrable on $[0, b]$?

- (c) Let f be a bounded function on $[a, b]$. If P and Q are partitions of $[a, b]$ such that $P \subseteq Q$, where Q has exactly one point extra than the points of P , then $L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P)$.

3. (a) Show that the improper integral $\int_1^{\infty} \frac{\sin t}{t} dt$ converges but not absolutely convergent.

- (b) Show that the improper integral $\int_0^1 x^m dx$ converges if and only if $m > -1$.

- (c) Examine the convergence of the following integrals:

(c) Show that the sequence $\langle x^2 e^{-ax} \rangle$ on $[0, \infty[$.

6. (a) Let $S(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ and $C(x)$ $x \in \mathbb{R}$. Prove that:

(i) $S' = C$ and $C' = -S$

(ii) $(S^2 + C^2)' = 0$

(iii) $S^2 + C^2 = 1$.

(b) Show that the function $E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ defined and is differentiable on $\mathbb{R}$ w

$E'(x) = E(x)$ for all $x \in \mathbb{R}$ and $E(x)$ all $x, y \in \mathbb{R}$.

(c) Find the radius of convergence of the series:

(i) $\sum_{n=0}^{\infty} \frac{(n!)^2}{(2n)!} x^n$

(ii) $\sum_{n=1}^{\infty} n^{-\sqrt{n}} x^n$